Errata

(Mathematical Introduction to Data Science by Sven A. Wegner)

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- Page 8, Line -4:
$$\sum_{i=1}^{n}(x_i-\overline{x})(y_i-\overline{y}) = \sum_{i=1}^{n} x_i y_i - \overline{y}\sum_{i=1}^{n} x_i - \overline{x}\sum_{i=1}^{n} y_i + n\overline{x}\,\overline{y}$$
- Page 9, Line -3:
$$0 = \sum_{i=1}^{n}(ax_i+b-y_i) = a\sum_{i=1}^{n} x_i + nb - \sum_{i=1}^{n} y_i = an\overline{x} + nb - n\overline{y},$$
- Page 11, Line 8: $\mathrm{P}\big[|Y_i(f)-y_i|<\delta\big] = \int_{y_i-\delta}^{y_i+\delta} \frac{1}{\sqrt{2\pi}\sigma}\,\mathrm{e}^{-\frac{(t-f(x_i))^2}{2\sigma^2}}\,\mathrm{d}t$
- Page 11, Line 10f:
$$\mathrm{P}\big[|Y_i(f)-y_i|<\delta \text{ für alle } i\big] = \int_{Q_\delta(y)} \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma}\,\mathrm{e}^{-\frac{(f(x_i)-t_i)^2}{2\sigma^2}}\,\mathrm{d}(t_1,\dots,t_n),$$
if we abbreviate the cube around $y=(y_1,\dots,y_n)\in\mathbb{R}^n$ with edge length 2δ by $Q_\delta(y) = [y_1-\delta, y_1+\delta]\times\cdots\times[y_n-\delta, y_n+\delta]$.
- Page 13, Line 10: …constant random variable av.
- Page 13, Line -5: since $\overline{x^{(n)2}} = \mathrm{var}(x^{(n)}) + \overline{x^{(n)}}^2$ as the sum …
- Page 15, Line -9: $\cdots + \frac{2}{n^2}\sum_{i<j} x_i x_j\,\mathrm{E}(\mathcal{E}_i)\,\mathrm{E}\,\mathcal{E}_j)$
- Page 16, Line 10: …, i.e., $f^*(x) = \langle a^*, x\rangle + b^*$ …
- Page 14, Line -2: If the latter is the case, then $\mathrm{sign}(r_{xy}) = \mathrm{sign}(\langle u, v\rangle) = \cdots$
- Page 17, Line 13: …we calculate (with just for now $\phi(\tilde{a}) = \langle \tilde{a}, X^{\mathrm{T}}X\tilde{a}\rangle$):
- Page 23, Line 9: In the picture it must be the z-axis.
- Page 23, Line -3: …and $(w,b) = (w_1,\dots,w_d,b)$ for …
- Page 24, Line 16: $\mathrm{P}\big[Y_i\cancel{(f)} = y_i \text{ for all } i\big] = \cdots$
- Page 24, Line 18: $L\colon \big\{f\colon \mathbb{R}^d \to (0,1) \mid f \text{ logistic function}\big\} \to \mathbb{R}$
- Page 26, Line 13: $h'(t) = \mathrm{sig}(t)\cancel{+1}$
- Page 27, Line 18: …the data *overlaps* if for every $w\in\mathbb{R}^{d+1}\setminus\{0\}$ there exists …
- Page 27, Line 22: …, the rounded ~~logisitic~~ logistic regressor, …
- Page 27, Line -10: if $\mathrm{sig}(\langle w, \widehat{x}\rangle) \geqslant 1/2$
- Page 33, Line 6: $\nabla\ell(x) = \sum_{i=1}^{n}\big(\mathrm{sig}(\langle w, \widehat{x}_i\rangle) - y_i\big)\widehat{x}_i$
- Page 37, Line 1:

```
function k-NN Classifier(D, k, x)
    D' ← D,  A ← ∅
    for j ← 1 to k do
        z* ← argmin_{z∈D'} ρ(x, π_1(z))
        A ← A ∪ {z*},  D' ← D' \ {z*}
    for y in Y do
        N(y) ← #{a ∈ A | π_2(a) = y}
    ℓ ← argmax_{y∈Y} N(y)
    return ℓ
```

Here, $\pi_2(x,y) = y$ denotes the projection onto the second entry of $(x,y)\in D$ ~~and y^* is the label of z^*~~.

- Page 39, Line 9: *... and* $k < n$.
- Page 39, Line 10: *The calculation of the k-neareast neighbors of x can be implemented such that at most $(n \cdot d \cdot k)$-many mulitplikations have to be carried out.*
- Page 39, Line 14: In the Euklidean metric, it requires $(d-1)$-many multiplications to compute one distance if we omit the root, which we can do as it does not change the argmin. This leads to

$$(d-1) \cdot (n + (n-1) + \cdots + (n-k+1)) \leqslant \cancel{C} \, d \cdot k \cdot n$$

  multiplikationen ~~with a suitable $C \in \mathbb{N}$~~.
- Page 40, Line 3: ..., we choose k-nearest neighbors $x_1, \ldots, x_k$ of x and denote their labels by $y_1, \ldots, y_k$.
- Page 40, Line 9: $f \colon X \to Y,\ f(x) = \dfrac{\sum_{i=1}^{k} w(x_i, x) \cdot y_i}{\sum_{i=1}^{k} w(x_i, x)}$
- Page 41, Line 5: $\tilde{x}^{(i)} = \left(a + \dfrac{(x_1^{(i)} - \min_{j=1,\ldots,n} x_1^{(j)})(b-a)}{\max_{j=1,\ldots,n} x_1^{(j)} - \min_{j=1,\ldots,n} x_1^{(j)}}, \ldots \right)$
- Page 41, Line 7: $\tilde{x}^{(i)} = \left(\dfrac{x_1^{(i)} - \overline{x_1^{(\cdot)}}}{\sigma_1}, \ldots \right)$
- Page 42, Line 14: $\rho(x^{(1)}, x^{(4)}) = 3.681$
- Page 46, Line 1: We discuss some of these methods in Exercise 3.10.
- Page 45, Line 28: We thus see that texts no. 1 and text no. 2 are significantly more cosine similar than text no. 1 and text no. 3 or text no. 2 and text no. 3.
- Page 48, Line 26: The cosine distance, on the other hand, may appear here more natural, as the scalar product increases if the frequency of the fixed word increases in the second text.
- Page 49, Line 16 – 17: (i) What ratings for Product 4 by Customer 5 arise from the method explained in Example 3.12 and which from the method explained in Example 3.19? — When applying the method, a problem arises with Customer 2; consider whether this problem is likely to occur frequently real rating tables and what a sensible way to deal with it might be.
- Page 52, Line 11: For finite subsets $A, B \subseteq X$ we define ...
- Page 53, Line 20:

```
function LINKAGE-BASED CLUSTERING (X, ρ, D, δ)
    k ← #D
    for i ← 1 to k do
        C_i ← {x_i}
    while min_{i≠j} ρ(C_i, C_j) ⩽ δ and k ⩾ 2 do
        m ← 0
        (i*, j*) ← argmin_{i≠j} ρ(C_i, C_j)
        for ℓ ← 1 to k − 1 do
            if ℓ = min(i*, j*) then
                C_ℓ ← C_{i*} ∪ C_{j*}
            if ℓ = max(i*, j*) then
                m ← 1
                C_ℓ ← C_{ℓ+m}
            else
                C_ℓ ← C_{ℓ+m}
        k ← k − 1
    return C_1, ..., C_k
```

- Page 54, Line −6: $K \colon \mathcal{C}_k \to \mathbb{R}$

- Page 56, Line −16: *The following pseudocode approximates a minimizer of the k-means cost function.*
- Page 56, Pseudocode:

```
1:  function K-MEANS (D, k, X, ρ)
2:      μ_1, ..., μ_k ← pairwise different points from X
3:      for i ← 1 to k do
4:          C_i ← {x ∈ D | i ∈ argmin_{j=1,...,k} ρ(x, μ_j)}
5:      U ← True
6:      while U = True do
7:          U ← False
8:          for i ← 1 to k do
9:              μ_i ← μ(C_i)
10:         for i ← 1 to k do
11:             C'_i ← {x ∈ D | i ∈ argmin_{j=1,...,k} ρ(x, μ_j)}
12:             if C'_i ≠ C_i
13:                 C_i ← C'_i
14:                 U ←True
15:     return C_1, ..., C_k
```

  In the lines 4 and 9 of the pseudocode we pick as a single i in the case that the armin is not unique.
- Page 57, Line 7:

$$\mu(A) \in \operatorname*{argmin}_{\mu\in A} \sum_{x\in A} \rho(x,\mu)^2, \quad \text{respectively} \quad \mu(A) \in \operatorname*{argmin}_{\mu\in X} \sum_{x\in A} \rho(x,\mu).$$

- Page 57, Line −7: For $j \geqslant 1$ denote by $(C_1^{(j)}, \dots, C_k^{(j)})$ that clustering which the algorithm produces in the j-th round. For $j \geqslant 2$ we have

$$K(C_1^{(j)}, \dots, C_k^{(j)}) \quad = \min_{\mu_1,\dots,\mu_k\in X} \sum_{i=1}^{k} \sum_{x\in C_i^{(j)}} \rho(x,\mu_i)^2$$

- Page 58, Line 3: ...line 11 of Algorithm 4.9 ...
- Page 58, Line 7: There, Picture 1b corresponds to the penultimate line in the estimate and Picture 2a to the line above that.
- Page 58, Line 10: ...as we have just moved point x_3 from cluster C_2 in Figure 1b to cluster C_1 in Figure 2a, ...
- Page 59, Line −4: We assume that we start with the initial values $\mu_1 = 2$ and ...
- Page 62, Line 11: $A = (a_{ij})_{i,j=1,\dots,n}$
- Page 62, Line 13: $L = (\ell_{ij})_{i,j=1,\dots,n}$
- Page 63, Line −2: ...In Example 5.7, $\lambda_2 \neq 0$ and there are no clusters (or, depending on how one prefers to see it, one single cluster), in Example 5.8 ...
- Page 67, Line 8: For the other direction let $\{v_1, \dots, v_n\}$ be a basis consisting of eigenvectors corresponding to the λ_i and let $U \subseteq \mathbb{R}^n$ be a subspace with $\dim U = n-k+1$. By construction $U \cap \operatorname{span}\{v_1, \dots, v_k\} \neq \{0\}$ and we can select $0 \neq x = \alpha_1 v_1 + \dots + \alpha_k v_k \in U$. Then it follows

$$\frac{\langle x, Mx\rangle}{\langle x, x\rangle} = \frac{\sum_{i=1}^{k} \lambda_i \alpha_i^2}{\sum_{i=1}^{k} \alpha_i^2} \leqslant \frac{\lambda_k \sum_{i=1}^{k} \alpha_i^2}{\sum_{i=1}^{k} \alpha_i^2} = \lambda_k,$$

  since the λ_i's are increasing. With this we get $\min_{0\neq x\in U} \frac{\langle x,Mx\rangle}{\langle x,x\rangle} \leqslant \lambda_k$ which then leads to

$$\max_{\substack{U\subseteq\mathbb{R}^n \\ \dim U = n-k+1}} \min_{\substack{x\in U \\ x\neq 0}} \frac{\langle x, Mx\rangle}{\langle x, x\rangle} \leqslant \lambda_k.$$

- Page 67, Line -4: ...and any corresponding eigenvector v_2 of norm 1:
- Page 68, Line 14: Let $G = (V, E)$ be a graph with $\deg(v) > 0$ for all $v \in V$.
- Page 70, Line 10:

$$\lambda_2(\mathcal{L}) = \min_{\substack{x \neq 0 \\ \langle Dx, \mathbb{1} \rangle = 0}} \frac{\sum_{\{i,j\} \in E} (x_i - x_j)^2}{\sum_{i=1}^{n} x_i^2 d_i}.$$

- Page 70, Line 16:

$$\mathcal{L} D^{1/2} \mathbb{1} = D^{-1/2} L D^{-1/2} D^{1/2} \mathbb{1} = D^{-1/2} L \mathbb{1} \underset{\substack{\uparrow \\ \text{Proposition} \\ 5.6}}{=} D^{-1/2} 0 \mathbb{1} = 0 D^{1/2} \mathbb{1}.$$

- Page 70, Line 18:

$$\begin{aligned} \lambda_2(\mathcal{L}) &\overset{\substack{\text{Theorem} \\ 5.13}}{=} \min_{\substack{x \neq 0 \\ \langle x, D^{1/2}\mathbb{1} \rangle = 0}} \frac{\langle x, \mathcal{L} x \rangle}{\langle x, x \rangle} \overset{(*)}{=} \min_{\substack{y \neq 0 \\ \langle D^{1/2} y, D^{1/2} \mathbb{1} \rangle = 0}} \frac{\langle D^{1/2} y, \mathcal{L} D^{1/2} y \rangle}{\langle D^{1/2} y, D^{1/2} y \rangle} \\ &= \min_{\substack{y \neq 0 \\ \langle y, D\mathbb{1} \rangle = 0}} \frac{\langle y, L y \rangle}{\langle y, D y \rangle} = \min_{\substack{y \neq 0 \\ \langle Dy, \mathbb{1} \rangle = 0}} \frac{\sum_{\{i,j\} \in E} (y_i - y_j)^2}{\sum_{i=1}^{n} y_i^2 d_i} \end{aligned}$$

- Page 71, Line 14: *(ii)* $\min(\operatorname{vol} S_k, \operatorname{vol} S_k^c) = \operatorname{vol} S_k^c$ *and* $\operatorname{vol} S_k^c - \operatorname{vol} S_{k+1}^c = d_{k+1}$ *hold whenever* $r \leqslant k \leqslant n - 1$.
- Page 71, Line 17:

$$\operatorname{vol} S_k^c - \operatorname{vol} S_{k+1}^c = \sum_{i=k+1}^{n} d_i - \sum_{i=k+2}^{n} d_i = d_{k+1}$$

- Page 72, Line 9:

$$\langle Dx, \mathbb{1} \rangle = \sum_{i=1}^{n} d_i x_i = \dots$$

- Page 72, Line -1 (and Page 73, Line 1):

$$\begin{aligned} \dots &= \operatorname{vol} S - \operatorname{vol} S \cdot \frac{\operatorname{vol} S}{\operatorname{vol} S + \operatorname{vol} S^c} \\ &\geqslant \operatorname{vol} S - \operatorname{vol} S \cdot \frac{\operatorname{vol} S}{2 \operatorname{vol} S}, \end{aligned}$$

- Page 73, Line 9: ...and our goal in the following will be to show $\lambda_2 \geqslant \alpha^2/2$, ...
- Page 73, Line 12 (Equation (5.2)):

$$\cdots \text{ and } \langle Dx, \mathbb{1} \rangle = \sum_{i=1}^{n} d_i x_i = 0$$

- Page 73, Line 21:

$$\begin{bmatrix} x_1 - x_r \\ \vdots \\ x_{r-1} - x_r \\ 0 \\ x_{r+1} - x_r \\ \vdots \\ x_n - x_r \end{bmatrix} = \begin{bmatrix} x_1 - x_r \\ \vdots \\ x_{r-1} - x_r \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ x_r - x_{r+1} \\ \vdots \\ x_r - x_n \end{bmatrix} =: p - n.$$

- Page 74, Line -3 (until top of page 75):

$$\lambda_2 = \frac{\sum_{\{i,j\} \in E} (x_i - x_j)^2}{\sum_{i=1}^{n} x_i^2 d_i}$$

$$
\begin{aligned}
&\underset{\substack{\uparrow\\(5.3)\\(5.4)}}{\geqslant} \frac{\sum_{\{i,j\}\in E}\left((p_i-p_j)^2+(n_i-n_j)^2\right)}{\sum_{i=1}^n (p_i^2+n_i^2)d_i}\\
&= \frac{\sum_{\{i,j\}\in E}(p_i-p_j)^2+\sum_{\{i,j\}\in E}(n_i-n_j)^2}{\sum_{i=1}^n p_i^2 d_i+\sum_{i=1}^n n_i^2 d_i}\\
&\underset{\substack{\uparrow\\(5.5)}}{\geqslant} \min\left(\frac{\sum_{\{i,j\}\in E}(p_i-p_j)^2}{\sum_{i=1}^n p_i^2 d_i},\frac{\sum_{\{i,j\}\in E}(n_i-n_j)^2}{\sum_{i=1}^n n_i^2 d_i}\right)\\
&= \min\left(\frac{\sum_{\{i,j\}\in E}(p_i-p_j)^2}{\sum_{i=1}^n p_i^2 d_i}\cdot\frac{\sum_{\{i,j\}\in E}(p_i+p_j)^2}{\sum_{\{i,j\}\in E}(p_i+p_j)^2},\dots\right)\\
&=: \min\left(\frac{Z}{N},\dots\right).
\end{aligned}
$$

- Page 75, Line 8:

$$
N=\sum_{i=1}^n p_i^2 d_i\cdot\sum_{\{i,j\}\in E}(p_i+p_j)^2 \underset{\substack{\uparrow\\(5.5)}}{\geqslant} \sum_{i=1}^n p_i^2 d_i\cdot\sum_{\{i,j\}\in E}2(p_i^2+p_j^2)
$$

- Page 76, Line 3:

$$
\cdots \underset{\substack{\uparrow\\\text{telescopic}\\\text{sum}}}{=} \left(\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\mathbb{1}_E(i,j)\sum_{k=i}^{j-1}\left(p_k^2-p_{k+1}^2\right)\right)^2
$$

- Page 79, Line 13: Finally, we want to note that Theorem 5.21 together with Remark 5.17(i) provides an upper bound for the eigenvalue $\lambda_2(\mathcal{L})$.
- Page 82, Line -5: *Let a dataset* $D=\{x_1,\dots,x_n\}\subseteq\mathbb{R}^d$ *be given, ...*
- Page 83, Line 3:

$$
\sum_{j=1}^k \|Xv_j\|^2 \geqslant \sum_{j=1}^k \|Xw_j\|^2
$$

- Page 83, Line -8:

$$
\{v_1,\dots,v_k\} \in \underset{\substack{\{\tilde v_1,\dots,\tilde v_k\}\subseteq\mathbb{R}^d\\\text{orthonormal}\\\text{system}}}{\operatorname{argmax}} \sum_{j=1}^n \|X\tilde v_j\|^2.
$$

- Page 84, Line 10: Let $D=\{x_1,\dots,x_n\}\subseteq\mathbb{R}^d$ be given ...
- Page 84, Line 13: for $j=1,\dots,k$, where ...
- Page 85, Line -1: $\langle w_{k+1},v_i\rangle=\langle w_{k+1},\underbrace{v_i-\pi_W(v_i)}_{\perp W}\rangle+\langle w_{k+1},\underbrace{\pi_W(v_i)}_{\in U}\rangle=0+0=0$
- Page 86, Line 11: Let $D=\{x_1,\dots,x_n\}\subseteq\mathbb{R}^d$ be given...
- Page 87, Line -9: Determine, by using Theorem 6.3 and a Python package for maximizations under nonlinear constraints, a 1-best fitting subspace for ...
- Page 86, Line 16: ...hold if $r=\dim\operatorname{span}D>0$. If $r=0$, then all σ_k vanish.
- Page 86, Line 19: For $k\geqslant r+1$ we apply the equation in the proof of Lemma 6.2 twice, on the one hand for V_k with ONB $\{v_1,\dots,v_k\}$, and on the other for V_{k-1} with ONB $\{v_1,\dots,v_{k-1}\}$. Then, after rearranging, we obtain

$$
\sigma_k^2=\|Xv_k\|^2=\sum_{i=1}^n \operatorname{dist}(x_i,V_{k-1})^2-\sum_{i=1}^n \operatorname{dist}(x_i,V_k)^2.
$$

  As $V_r\subseteq V_{k-1}\subseteq V_k$ holds, Lemma 6.4 implies that both terms on the right hand side are zero. Thus, $\sigma_{r+1},\dots,\sigma_d=0$.
- Page 90, Line 8: $\|u\|^2=\|Av\|^2=\langle Av,Av\rangle=\cdots$

- Page 94, Line 6: ..., it may be easier to start with this matrix and proceed analologuously (notice that one then gets the u_i's first and has to compute the v_i's by applying A^{T}).
- Page 94, Line -9: $V \in \mathbb{R}^{d\times d}$
- Page 94, Line -7: Then precisely r-many of the σ_i are nonzero and these are the singular values of A ...
- Page 96, Line 10: ...and let $\{v_1, \dots, v_r\}$ be some arbitrary orthonormal system ...
- Page 96, Line 14: By construction, the $v_1, \dots, v_r$ that we defined in the theorem form an orthonormal system.
- Page 96, Line 20:

$$\|Av\|^2 = \langle v, A^{\mathrm{T}}Av\rangle = \langle v, B\operatorname{diag}(\cdots)B^{\mathrm{T}}v\rangle = \langle B^{\mathrm{T}}v, \operatorname{diag}(\cdots)B^{\mathrm{T}}v\rangle = \sum_{j=1}^{r} \lambda_j^{\cancel{2}}(B^{\mathrm{T}}v)_j^2$$

  In the rest of the proof everywhere the squares at the eigenvalues λ_j have to be removed!
- Page 98, Line 4: ...v_i is therefore an eigenvector of $A^{\mathrm{T}}A$ corresponding to σ_i^2 for $i = 1, \dots, k$.
- Page 99, Line -3: ~~Multiplication with V^{T} from the left in~~
- Page 99, Line 12: $v = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \sigma_{k+1} \\ w_{k+1} \end{bmatrix} \Bigg/ \left\| \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \sigma_{k+1} \\ w_{k+1} \end{bmatrix} \right\|$
- Page 99, Line 14: ..., we want to discuss how the singular value decomposition ~~indeed~~ generalizes the orthogonal diagonalization theorem.
- Page 99, Line 16: **Proposition 7.14.** Let $A \in \mathbb{R}^{n\times n}$ be a symmetric and positive semi-definite matrix, ...
- Page 99, Line -3: Proposition 7.14 thus yields exactly an orthogonl diagonalization $V^{\mathrm{T}}AV = \operatorname{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0)$; but only under the additional assumption of A being positive semidefinite.

  Corollary 5.22. Let $G = (V, E)$ be a graph with $\deg(i) > 0$ for all $i \in V$. Then for the second smallest eigenvalue λ_2 of the normalized Laplace matrix of G the estimate $\lambda_2 \leqslant 2$ holds. □
- Page 102, Line 17: *Proof.* Let $B \in \mathbb{R}^{n\times d}$ be arbitrary. Put

$$V_k := \operatorname{span}\{v_1, \dots, v_k\} \quad \text{and} \quad W := \operatorname{span}\{b_1, \dots, b_n\} \quad \cancel{\text{and} \quad \check{V} := \operatorname{span}\{\check{a}_1, \dots, \check{a}_n\},}$$

  where $b_i \in \mathbb{R}^d$ is the vector that has as entries those of the i-th row of B. According to Theorem 7.12, V_k is a k-best fitting subspace for $\{a_1, \dots, a_n\}$. ~~We claim that $\check{V} = V_k$. By Theorem 7.17(ii), we have $\check{a}_i \in V_k$ for $i = 1, \dots, n$ and thus $\check{V} \subseteq V_k$. The equality then follows from $\dim \check{V} = \operatorname{rk} \check{A} = k = \dim V_k$.~~ Consequently ~~we obtain for each $i = 1, \dots, n$ the estimates~~

$$\begin{aligned}
\|A - B\|_{\mathrm{F}}^2 &= \sum_{i=1}^{n} \|a_i - b_i\|^2 \underset{\substack{\uparrow \\ b_i \in W}}{\geqslant} \sum_{i=1}^{n} \|a_i - \pi_W(a_i)\|^2 \underset{\substack{\uparrow \\ V_k\ k\text{-best} \\ \text{fitting for} \\ a_1,\dots,a_n}}{\geqslant} \sum_{i=1}^{n} \|a_i - \pi_{V_k}(a_i)\|^2 \\
&\underset{\substack{\uparrow \\ \text{Thm} \\ 7.17\text{(ii)}}}{=} \sum_{i=1}^{n} \|a_i - \check{a}_i\|^2 = \|A - \check{A}\|_{\mathrm{F}}^2
\end{aligned}$$

  which concludes the proof. □
- Page 105, Line -1 and Page 106, Line 1:

$$A = \underbrace{\begin{bmatrix} 0.07 & 0.29 & 0.32 & 0.51 & 0.66 & 0.18 & -0.23 \\ 0.13 & -0.02 & -0.01 & -0.79 & 0.59 & -0.02 & -0.06 \\ 0.68 & -0.11 & -0.05 & -0.05 & -0.24 & 0.56 & -0.35 \\ 0.15 & 0.59 & 0.65 & -0.25 & -0.33 & -0.09 & 0.11 \\ 0.41 & -0.07 & -0.03 & 0.10 & -0.02 & -0.78 & -0.43 \\ 0.07 & 0.73 & -0.67 & 0.00 & -0.00 & 0.00 & 0.00 \\ 0.55 & -0.09 & -0.04 & 0.17 & 0.17 & -0.11 & 0.78 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} 12.4 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 9.5 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 1.3 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} 0.56 & 0.09 & 0.56 & 0.09 & 0.59 \\ -0.12 & 0.69 & -0.12 & 0.69 & 0.02 \\ -0.40 & -0.09 & -0.40 & -0.09 & 0.80 \\ \cancel{0.41} & \cancel{0.09} & \cancel{0.40} & \cancel{0.09} & \cancel{0.80} \\ 0.51 & 0.48 & -0.51 & -0.48 & -0.00 \\ 0.48 & -0.51 & -0.48 & 0.51 & -0.00 \end{bmatrix}}_{V^{\mathrm{T}}}$$

$$= \begin{bmatrix} 0.07 & 0.29 & 0.32 \\ 0.13 & -0.02 & -0.01 \\ 0.68 & -0.11 & -0.05 \\ 0.15 & 0.59 & 0.65 \\ 0.41 & -0.07 & -0.03 \\ 0.07 & 0.73 & -0.67 \\ 0.55 & -0.09 & -0.04 \end{bmatrix} \begin{bmatrix} 12.4 & & \\ & 9.5 & \\ & & 1.3 \end{bmatrix} \begin{bmatrix} 0.56 & 0.09 & 0.56 & 0.09 & 0.59 \\ -0.12 & 0.69 & -0.12 & 0.69 & 0.02 \\ -0.40 & -0.09 & -0.40 & -0.09 & 0.80 \end{bmatrix}.$$

- Page 106, Line 6:

$$\check{A} = \begin{bmatrix} 0.15 & 1.97 & 0.15 & 1.97 & 0.56 \\ 0.92 & 0.01 & 0.92 & 0.01 & 0.94 \\ 4.84 & 0.03 & 4.84 & 0.03 & 4.95 \\ 0.36 & 4.03 & 0.36 & 4.03 & 1.20 \\ 2.92 & -0.00 & 2.92 & -0.00 & 2.98 \\ -0.34 & 4.86 & -0.34 & 4.86 & 0.65 \\ 3.92 & 0.02 & 3.92 & 0.02 & 4.00 \end{bmatrix} = \begin{bmatrix} 0.07 & 0.29 \\ 0.13 & -0.02 \\ 0.68 & -0.11 \\ 0.15 & 0.59 \\ 0.41 & -0.07 \\ 0.07 & 0.73 \\ 0.55 & -0.09 \end{bmatrix} \begin{bmatrix} 12.4 & \\ & 9.5 \end{bmatrix} \begin{bmatrix} 0.56 & 0.09 & 0.56 & 0.09 & 0.59 \\ -0.12 & 0.69 & -0.12 & 0.69 & 0.02 \end{bmatrix}$$

- Page 107, Line 15:

$$= \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} 0.07 & 0.29 & \cdots & -0.23 \\ 0.13 & -0.02 & & -0.06 \\ 0.68 & -0.11 & & -0.35 \\ 0.15 & 0.59 & & 0.11 \\ 0.41 & -0.07 & & -0.43 \\ 0.07 & 0.73 & & 0.00 \\ 0.55 & -0.09 & \cdots & 0.78 \end{bmatrix} \begin{bmatrix} 12.4 & & & \\ & 9.5 & & \\ & & 1.3 & \\ & & & \ddots \end{bmatrix} \begin{bmatrix} 0.56 & 0.09 & 0.56 & 0.09 & 0.59 \\ -0.12 & 0.69 & -0.12 & 0.69 & 0.02 \\ \vdots & & & & \vdots \\ 0.48 & -0.51 & -0.48 & 0.51 & 0.00 \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

- Page 107, Line -2:

$$u_2 = \quad 0.29 \cdot \text{Abbie} - 0.02 \cdot \text{Bailey} + \cdots - 0.09 \cdot \text{Gladys},$$

- Page 109, Line 10:

$$\begin{array}{r} \text{Abbie} \\ \text{Bailey} \\ \text{Catherine} \\ \text{Darlene} \\ \\ \\ \text{Elena} \\ \text{Fatima} \\ \text{Gladys} \end{array} \quad \check{A} = \begin{bmatrix} 0.07 & 0.29 \\ 0.13 & -0.02 \\ 0.68 & -0.11 \\ 0.15 & 0.59 \\ 0.41 & -0.07 \\ 0.07 & 0.73 \\ 0.55 & -0.09 \end{bmatrix} \begin{bmatrix} 12.4 & \\ & 9.5 \end{bmatrix} \begin{array}{c} \begin{array}{ccccc} \text{Alien} & \text{Casablanca} & \text{Star Wars} & \text{Titanic} & \text{The Matrix} \end{array} \\ \begin{bmatrix} 0.56 & 0.09 & 0.56 & 0.09 & 0.59 \\ -0.12 & 0.69 & -0.12 & 0.69 & 0.02 \end{bmatrix} \end{array}$$

- Page 109, Line -3: ... and $\check{\mathcal{V}} = \{v_1, v_2\}$.
- Page 119, Line -7 ..., we assume that the distribution has mean zero and variance 1.
- Page 120, Line 1 in eq. (8.2): $V(\|X\|^2) = V(X_1^2) + \cdots + V(X_d^2)$.
- Page 121, Line 4: ..., we get $\mathrm{E}(S_d) = 0$
- Page 121, Line 13: $= 2\sqrt{d}\,\mathrm{E}(\|X\| - \sqrt{d}\,)$
- Page 121, Line -2: **Theorem 8.2.** Let $X, Y \sim \mathcal{N}(0, 1, \mathbb{R}^d)$ be independent. Then *(i)* $\forall\, d \in \mathbb{N}$: $|\,\mathrm{E}(\|X - Y\| - \sqrt{2d})| \leqslant \sqrt{2/d}$,
- Page 122, Line 1: *(ii)* $\forall\, d \in \mathbb{N}$: $\mathrm{V}(\|X - Y\|) \leqslant 4$.
- Page 122, Line 3 in eq. (8.2): $d \cdot \big(\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} x^4 \exp(-x^2/2)\,\mathrm{d}x - \mathrm{V}(X_1)^2\big)$
- Page 125, Line -8: *(i)* $\forall\, d \in \mathbb{N}$: $|\,\mathrm{E}(\|X - Y\| - \sqrt{2d})| \leqslant \sqrt{2/d}$,
- Page 125, Line -7: *(ii)* $\forall\, d \in \mathbb{N}$: $\mathrm{V}(\|X - Y\|) \leqslant 4$.
- Page 125, Line -6: Firstly, calculate $\mathrm{V}((X_i - Y_i)^2) = 8$ ...
- Page 125, Line -5: Then infer $\mathrm{V}((X_i - Y_i)^2) = 8d$ ...
- Page 140, Line -7: *Proof.* We put $Y_1 + \cdots + Y_d$ and ...
- Page 140, last expression in eq. (10.1): $= \mathrm{e}^{-ta} \prod_{i=1}^{d} \mathrm{E}(\mathrm{e}^{tY_i})$

- Page 143, Lines 2 – 6:

$$\begin{aligned}
\mathrm{P}\big[\,\big|\|X\| - \sqrt{d}\,\big| > \varepsilon\big] &\leqslant \mathrm{P}\big[\,\big|\|X\| - \sqrt{d}\big| \cdot (\|X\| + \sqrt{d}) \geqslant \varepsilon \cdot \sqrt{d}\big] \\
&= \mathrm{P}\big[\,\big|X_1^2 + \cdots + X_d^2 - d\big| \geqslant \varepsilon\sqrt{d}\big] \\
&= \mathrm{P}\big[\,\big|(X_1^2 - 1) + \cdots + (X_d^2 - 1)\big| \geqslant \varepsilon\sqrt{d}\big] \\
&= \mathrm{P}\big[\,\big|\tfrac{X_1^2-1}{2} + \cdots + \tfrac{X_d^2-1}{2}\big| \geqslant \tfrac{\varepsilon\sqrt{d}}{2}\big] \\
&= \mathrm{P}\big[\,\big|Y_1 + \cdots + Y_d\big| \geqslant a\big],
\end{aligned}$$

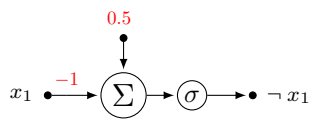
- Page 143, Line 10: 1 The Y_i are ~~pairwise~~ independent according to Fact 10.1(i).
- Page 152, Line −11:

$$\mathrm{P}\left[\,\left|\left\langle \frac{u_i}{\|u_i\|}, \frac{u_j}{\|u_j\|}\right\rangle\right| \leqslant \varepsilon\right] \geqslant 1 - \frac{2/\varepsilon + 7}{\sqrt{d}}.$$

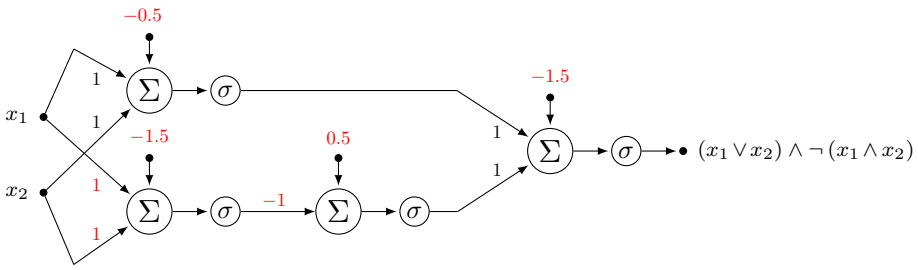
- Page 155, Line 5: ... (i) The dimension d of the ambient space ...
- Page 155, Line −3ff:

$$\begin{aligned}
\mathrm{P}\big[\cdots\big] &= 1 - \mathrm{P}\big[\,\exists\, i \neq j\colon \|T_U x_i - T_U x_j\| \notin \big((1-\varepsilon)\|x_i - x_j\|, (1+\varepsilon)\|x_i - x_j\|\big)\big] \\
&\geqslant 1 - \sum_{i \neq j}(1 - P_{ij}) \geqslant 1 - \sum_{i \neq j} 2\,\mathrm{e}^{-k\varepsilon^2/16} \geqslant 1 - \frac{n^2}{2} \cdot 2\,\mathrm{e}^{-k\varepsilon^2/16} \\
&\geqslant 1 - n^2\,\mathrm{e}^{-(\frac{48}{\varepsilon^2}\log n)\,\varepsilon^2/16} = 1 - n^2\,\mathrm{e}^{\log(n^{-3})} \\
&= 1 - \frac{1}{n}
\end{aligned}$$

- Page 169, Line −1: $\sqrt{(\Delta(d))^2 + 2d} - \sqrt{2d} \to \infty$
- Page 170, Line −1: $\lim_{d\to\infty} \sqrt{(\Delta(d))^2 + 2d} - \sqrt{2d} = \infty$
- Page 171, Line 4: $\sqrt{(\Delta(d))^2 + 2d} - \sqrt{2d} \geqslant \sqrt{R\sqrt{d} + 2d} - \sqrt{2d} \xrightarrow{d\to\infty} \frac{R}{2\sqrt{2}}$
- Page 171, Line 5: Multiplication with $\sqrt{(\Delta(d))^2 + 2d} + \sqrt{2d}$ ...
- Page 173, Line 6: $\cdots \geqslant R \cdot \big(\sqrt{(\Delta(d))^2 + 2d} - \sqrt{2d}\,\big) \geqslant \cdots$
- Page 184, Line −6: $y_i\langle w^{(j+1)}, \widehat{x}_i\rangle = y_i\langle w^{(j)} + y_i\widehat{x}_i, \widehat{x}_i\rangle = \underbrace{y_i\langle w^{(j)}, \widehat{x}_i\rangle}_{\substack{\text{erroneously}\\ \text{less than zero}}} + \underbrace{y_i^2\,\langle \widehat{x}_i, \widehat{x}_i\rangle}_{\substack{\text{addition of a}\\ \text{positive cor-}\\ \text{rection term}}}.$
- Page 185, Line 18: Since D is linearly separable, there exists some $w \in \mathbb{R}^{d+1}\setminus\{0\}$, such that all points will be classified correctly. ~~This vector w cannot be the zero vector, because then $h \equiv 0$ and all data points would be classified incorrectly.~~
- Page 185, Line −4: $\langle w^*, w^{(j+1)}\rangle \geqslant \langle w^*, w^{(0)}\rangle + (j+1)\cdot\gamma = (j+1)\cdot\gamma$
- Page 186, Line 3: $\|w^{(j+1)}\|^2 \leqslant \|w^{(0)}\|^2 + (j+1)\cdot R^2 = (j+1)\cdot R^2$.
- Page 186, Line 3: $(j+1)\cdot\gamma\langle w^*, w^{(j+1)}\rangle \leqslant \|w^*\|^{\cancel{2}}\|w^{(j+1)}\|^{\cancel{2}} = \|w^{(j+1)}\|^{\cancel{2}} + \sqrt{j+1}\cdot R$
- Page 186, Line 1: under the underbrace: $\leqslant 0$, as ...
- Page 187, Line −8: Assume that the perceptron algorithm outputs the generalized weight vector ...
- Page 190, Line 2: and ~~let~~ $b \in \mathbb{R}$
- Page 227, Line 11:



- Page 227, Line −3:



- Page 229, Line 13: Alternatively, with sigmoid activation, ...
- Page 229, Line −4: $N = \begin{bmatrix} a_{11} & \cdots & a_{1\ell} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{m\ell} \end{bmatrix} \begin{bmatrix} n_1 \\ \vdots \\ n_\ell \end{bmatrix}$
- Page 229, Line −3: ...and a matrix $(a_{ij})_{i,j} \in \mathbb{R}^{m\times\ell}$ ...
- Page 236, Line −4: ...assume that σ has finite limits ...
- Page 237, Line 5: $\forall\, w \in \mathbb{R}^{\cancel{n}},\ b \in \mathbb{R}\colon \displaystyle\int_\Omega \sigma(wx+b) + \ell \,\mathrm{d}\mu(x) = 0$
- Page 237, Line −7: $\forall\, w \in \mathbb{R}^{\cancel{n}},\ b \in \mathbb{R}\colon \displaystyle\int_\Omega \sigma(wx+b) \,\mathrm{d}\mu(x) = 0$
- Page 242, Line −4: Since Ω is compact there exists $c_0 \geqslant 0$ such that $a_1 f_1 + \cdots + a_i f_i + c_0 \geqslant 0$ on Ω for all $i = 1, \ldots, \ell$. We construct ...
- Page 248, Line 1: ...we get the far left formula in the proposition. To prove the other two formulas, we use ...
- Page 282, Line 6: (ii) For A, $B \in \Sigma$ with $\mathrm{P}(B) \neq 0$, $\mathrm{P}(A|B) := \dfrac{\mathrm{P}(A\cap B)}{\mathrm{P}(B)}$ ...
- Page 283, Line 19:

$$\rho(x) = \tfrac{1}{(2\pi\sigma^2)^{d/2}}\, \mathrm{e}^{-\frac{\|x-\mu\|^2}{2\sigma^2}} \quad \text{respectively} \quad \rho(x) = \tfrac{1}{\lambda^d(B)} \cdot \mathbb{1}_B(x),$$

- Page 286, Line −5: For $A = A_1 \times \cdots \times A_d \subseteq \mathbb{R}^d$ with $A_i \in \mathcal{B}^{\cancel{d}}$ we calculate
- Page 288, Line 2:

$$\begin{aligned}
(\rho_1 * \rho_2)(s) &= \frac{1}{2\pi\sqrt{ab}} \int_{\mathbb{R}} \exp\Big(-\frac{(s-t)^2}{2a}\Big) \exp\Big(-\frac{t^2}{2b}\Big)\,\mathrm{d}t \\
&= \frac{1}{2\pi\sqrt{ab}} \int_{\mathbb{R}} \exp\Big(-\frac{b(s^2-2st+t^2)+at^2}{2ab}\Big)\,\mathrm{d}t \\
&= \frac{1}{2\pi\sqrt{ab}} \int_{\mathbb{R}} \exp\Big(-\frac{t^2(b+a)-2stb+bs^2}{2ab}\Big)\,\mathrm{d}t \\
&= \frac{1}{2\pi\sqrt{ab}} \int_{\mathbb{R}} \exp\Big(-\frac{t^2(b+a)/c-2stb/c+bs^2/c}{2ab/c}\Big)\,\mathrm{d}t \\
&= \frac{1}{\sqrt{2\pi c}\sqrt{2\pi(ab/c)}} \int_{\mathbb{R}} \exp\Big(-\frac{(t-(bs)/c)^2-(sb/c)^2+s^2(b/c)}{2ab/c}\Big)\,\mathrm{d}t \\
&= \frac{1}{\sqrt{2\pi c}} \exp\Big(+\frac{(sb/c)^2-s^2(b/c)}{2ab/c}\Big) \frac{1}{\sqrt{2\pi(ab/c)}} \int_{\mathbb{R}} \exp\Big(-\frac{(t-(bs)/c)^2}{2ab/c}\Big)\,\mathrm{d}t \\
&= \frac{1}{\sqrt{2\pi c}} \exp\Big(+\frac{(sb/c)^2c^2-s^2(b/c)c^2}{2abc}\Big) \\
&= \frac{1}{\sqrt{2\pi c}} \exp\Big(+\frac{s^2(b^2-bc)}{2abc}\Big) \\
&= \frac{1}{\sqrt{2\pi c}} \exp\Big(-\frac{s^2}{2c}\Big),
\end{aligned}$$